

Utkarsh Singh

+91 9871335300 | utkarsh.yash@gmail.com | linkedin.com/in/utkarsh-singh06 | github.com/UtkarshSingh-06 | leetcode.com/u/Utkarsh-Singh06

Skills

- **Languages:** C/C++, JavaScript (ES6+), TypeScript, Python, SQL
- **Backend:** Node.js, Express.js, RESTful APIs, Microservices, Authentication & Authorization (JWT, OAuth2, RBAC)
- **Frontend:** React.js, Next.js (App Router, SSR/ISR), HTML5, CSS3, Tailwind CSS
- **Databases:** MongoDB, MySQL, PostgreSQL, Database Design, Indexing & Query Optimization
- **Cloud & DevOps:** AWS, Azure, Docker, Kubernetes, Terraform, Ansible, CI/CD Pipelines (GitHub Actions)
- **Tools & Systems:** Git, GitHub, Linux, Kafka, Elasticsearch, Postman, VS Code, IntelliJ IDEA, PyCharm

Experience

SDE Intern

Jun 2026 – Present

Samsung Electro-Mechanics Software India (SEMB)

Bengaluru, India

- Developed and maintained full-stack enterprise applications using Python, JavaScript, and SQL, supporting manufacturing and quality-management workflows across multiple business teams.
- Designed and implemented RESTful APIs and backend services processing 10,000+ operational records, improving application scalability, responsiveness, and maintainability.
- Built analytics dashboards and reporting solutions that reduced manual tracking efforts by 60% and enabled faster operational decision-making.
- Automated data validation and report-generation workflows, improving execution efficiency and ensuring high data accuracy across internal systems.
- Collaborated with AI and QA teams throughout the SDLC to develop, test, debug, and deploy production-grade software, accelerating feature delivery and system reliability.

Head of Development

Sep. 2024 – May 2025

CampusAdda, Manipal University Jaipur

- Led development of a cross-platform campus application serving 10K+ daily active users across Android and iOS platforms.
- Architected and delivered production-ready features, improving platform scalability, user engagement, and overall application performance.
- Standardized development workflows and release processes, reducing deployment cycles by 30% and improving team productivity.

Open Source Contributor

August 2025 – November 2025

GirlScript Summer of Code (GSSoC '25)

- Contributed to production-grade open-source projects by implementing features, fixing bugs, and improving documentation.
- Collaborated via Git/GitHub workflows and code reviews, ensuring clean, maintainable, and scalable code across distributed teams.

Projects

Financial Fraud Detection & Risk Scoring Platform

Python, FastAPI, React, TensorFlow, AWS, Docker

- Built a real-time fraud detection system processing 10K+ transactions/day with 92%+ accuracy for automated risk scoring.
- Designed low-latency REST APIs (< 120 ms) using FastAPI and integrated with a React dashboard for real-time monitoring.
- Deployed containerized services on AWS using Docker, achieving 99.9% uptime with scalable infrastructure.

Enterprise AI Quality Control Platform

Python, Docker, Kubernetes, CI/CD, AWS, Terraform, Ansible

- Built an AI-driven quality control framework reducing manual QA effort by 60% through automated anomaly detection.
- Containerized microservices using Docker and orchestrated with Kubernetes, enabling scalable distributed workloads.
- Provisioned cloud infrastructure using Terraform and automated configuration using Ansible for consistent deployments.
- Designed CI/CD pipelines (GitHub Actions) reducing release time by 40%, and integrated monitoring systems (Prometheus, Grafana) ensuring 99.9% service reliability.

AI Security Operations Platform (AISecOps)

FastAPI, React, TypeScript, PostgreSQL, Redis, Kafka, Elasticsearch, Docker, Kubernetes, PyTorch

- Architected a multi-tenant cybersecurity platform processing 100K+ security events/day, leveraging event-driven microservices for real-time threat detection and risk analysis.
- Developed FastAPI services, Kafka pipelines, and ML models (XGBoost, PyTorch) achieving 92% accuracy in threat classification.
- Built an LLM-powered investigation assistant, RBAC-based incident management dashboard, and observability stack (Elasticsearch, Prometheus, Grafana), reducing investigation effort by 70% via automated log analysis.

Education

Manipal University Jaipur

Jaipur, India

B.Tech. Information & Technology

Aug 2023 – May 2027

Coursework: Data Structures, Algorithms, DBMS, Software Engineering, Computer Networks

Recipient of Multiple Student Excellence Awards by the Department;

Solved 400+ DSA problems on LeetCode and coding platforms

Certifications

- AWS Certified Solutions Architect - Associate (SAA-C03), Amazon Web Services
- NPTEL - Data Structures and Algorithms | Design and Analysis of Algorithms
- Red Hat System Administration I & II (RH124, RH134) - RHEL 9.3 | Oracle Academy - Database Programming with SQL